Management, Technology & Entrepreneurship

Principles of Finance

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Assignment 5

Instructions

- Assignments should be done in groups of 4 students.
- You should remain with the same group through the entire course.
- Submit on Moodle only one copy of solutions per group.
- For each assignment you can get a maximum of 100 points.
- All assignments turned in late will not be graded (zero points).
- Round off to two decimal places unless indicated otherwise.

Due date

The due date is indicated on Moodle.

For this assignment, you should hand in two files: First, a separate file (pdf) where you write all your solutions and record all your results (tables, graphs, etc.) and interpretations. Second, you should also hand in your Excel file with your calculations. The Excel file we only need if we do not understand something you did. The instructions are very precise on what you should do, what results to report, and what questions to answer.

Part I (45 points)

1. **(5 points)** You have collected the data about Bitcoin prices in the last 8 months:

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Price	38,483	43,193	45,539	37,715	31,792	19,785	23,337	20,050

   (a) What is the average montly return? What is the return over the period of seven month (Jan-Aug)?

   (b) What is the variance of the returns?

   (c) What is the standard deviation of the returns?

2. **(10 points)** Assume that the CAPM holds in the economy and that the risk-free rate is 3%. An expected return and volatility of the market portfolio are 11% and 10%, respectively. In addition, you have the following data about two portfolios, X and Y:

Portfolio	Volatility	Beta
X	0.15	1.5
Y	0.075	0.5

   Using this information, answer the following questions:

   (a) What is the expected returns of portfolio X? Portfolio Y?

   (b) What are the correlations of these two portfolios with the market portfolio? How do you interpret a correlation of 1?

   (c) Do these two assets lie on the SML? on the CML? Explain your answer.

   (d) Is Portfolio X a combination of the risk-free asset and the market portfolio? Can you say anything about the composition of Portfolio Y? Why?

3. **(15 points)** Suppose that the CAPM assumptions hold. The expected return of the market is 10% and its volatility is 15%. The risk-free rate is 1%. An investor is looking to invest in the following three stocks:

	ρ_{jM}	σ_j
A	0.3	35%
B	0.6	20%
C	-0.15	8%

   with $\rho_{j,M}$ denoting the correlation between the Market and asset j, and σ_j the volatility of asset j.

(a) Calculate the expected returns of the three stocks.

(b) Calculate the expected return of a portfolio that consists of 40% stock A, 40% stock B and 20% stock C.

(c) Calculate the beta of the portfolio in (b).

(d) Calculate the idiosyncratic risk of each stock.

(e) If the market is the only source of covariance between the three stocks, what will be the volatility of the portfolio in (b)? what are its idiosyncratic and systematic risks?

4. **(10 points)** Consider the following expected returns and variances:

	Expected Return	Variance
A	0.12	0.0154
B	0.17	0.0213

The covariance between the two returns is 0.008.

What is the composition of the minimum risk portfolio of these two shares? What is the expected return and risk of this portfolio?

5. **(5 points)** Assume that the CAPM is a good description of stock price returns. Consider an asset Z that generates a cash flow at $t = 1$ and does not pay anything at $t = 0$. The cash flow CF_1 is equal to:

$$CF_1 = a(1 + r_1^M)$$

where $a = 2.5$ and r_1^M is the realized return on market portfolio at $t = 1$.

The expected market return is equal 8%, and the risk-free rate is 2%.

What is the price of Asset Z at time $t = 0$?

Part II (55 points)

(20 points) You have just started your new job with a financial planning firm. In addition to studying for your exams, you have been asked to review a portion of a client's stock portfolio to determine the risk/return profiles of 12 stocks in the portfolio. Unfortunately, your small firm cannot afford the expensive databases that would provide all this information with a few simple keystrokes, but that is why they have you. Specifically, you have been asked to determine the monthly average returns and standard deviations for the 12 stocks. The stocks are: Apple (AAPL), Johnson & Johnson (JNJ), J. P. Morgan (JPM), Procter & Gamble (PG), Exxon Mobil (XOM), Pfizer Inc. (PFE), Microsoft Corp. (MSFT), AT & T (T), Citi Group (C), Oracle Corporation (ORCL), General Electric (GE), and Wells Fargo & Co. (WFC).

- Use the stock prices provided in the Excel file.
- Convert these prices to monthly returns as the percentage change in the monthly prices. *Hint: Pay attention to the chronological order of the data.*
- Compute the mean monthly returns and standard deviations for the monthly returns of each of the stocks. Convert the monthly statistics to **annual** statistics for easier interpretation. *Hint: The annualized return can be approximated by* $12\bar{r}_{monthly}$. *The annualized standard deviation can be approximated by* $\sqrt{12}\sigma_{monthly}$.
- Add a column in your Excel worksheet with the average returns across stocks for each month. These are the monthly returns to an equally weighted portfolio of the 12 stocks. Compute the mean and standard deviation of monthly returns for the equally weighted portfolio. Double check that the average return on this equally weighted portfolio is equal to the average return of all of the individual stocks. Again, convert these monthly statistics to annual statistics.

Report the following results:

1. Report the annual mean return and annualized standard deviation of the 12 stocks and of the equally-weighted portfolio in a table in the separate file.
2. Create a plot with the annual standard deviation on the x-axis and the annual return on the y-axis and represent the 12 stocks and the portfolio in this plot. Copy the plot into the separate file.
3. How does the portfolio average return and standard deviation compare with those of individual firms? Comment.

(35 points) Now you want to rebalance your portfolio with the optimal weights that will provide the best risk and return combination for the 12-stock portfolio. Use the Solver function in Excel to perform this analysis (the time-consuming alternative is to find the optimum weights by trial and error).

- Begin with the equally weighted portfolio analyzed above. Establish the portfolio returns for the stocks in the portfolio using a formula that depends on the portfolio weights. Initially, these weights will all equal 1/12. You would like to allow the portfolio weights to vary, so you will need to list the weights for each stock in separate cells and establish another cell that sums the weights of the stocks. The portfolio returns for each month MUST reference these weights for Excel solver to be of any use.

- Compute the values for the monthly mean return and standard deviation of the portfolio. Convert these values to annual numbers.

- Compute the efficient frontier when short sales are NOT allowed. To activate the Solver function in Excel, click the "Tools" tab, select "Add-Ins...", check "Solver-Add-in" in the pop-up dialog box, and then click "OK". To set the solver parameter:

  - Set the target cell as the cell of interest, making it the cell that computes the (annual) portfolio standard deviation. Minimize this value.
  - Establish the "By Changing Cells" by holding the control key and clicking in each of the 12 cells containing the weights of each stock (all in all you select 12 cells).
  - Add constraints by clicking on the add button next to the "Subject to the constraints" box. One set of constraints will be that the weight of each stock is greater than or equal to zero. A second constraint is that the weights will sum to one.
  - Compute the portfolio with the lowest standard deviation for a given expected return. Start by finding this portfolio with an expected return of 9%. To do this, add a constraint that the (annual) portfolio return equals 0.09.
  - If the parameters are set correctly, you should get a solution when you click "Solve". If there is an error, you will need to double-check the parameters, especially the constraints.

- Record the resulting standard deviation for the "optimally weighted" portfolio with a return of 0.09 in a separate cell on the spreadsheet. Repeat the previous step to solve for the portfolio with the lowest standard deviation for several different choices of expected returns: 0.11, 0.13, 0.15, 0.20, 0.25, 0.30, 0.40. Record these values. Plot the efficient frontier with the constraint of no short sales. To do this, use a plot with portfolio standard deviation on the x-axis and the return on the y-axis.

- Redo your analysis to allow for short sales by removing the constraint that each portfolio weight is greater than or equal to zero. Use again the solver to calculate the (annual) portfolio standard deviation when the annual portfolio returns are set to 0.09, 0.11, 0.13, 0.15, 0.20, 0.25, 0.30, 0.40. Plot the unconstrained efficient frontier into the same plot as the constrained efficient frontier.

Report the following results:

1. Report the values that you used to produce the plots of the constrained and unconstrained efficient frontier in a table in the separate file.

2. Copy the plot into the separate file.

3. What do you observe in the plot? Comment.